DERIVED COMPLETION COMMUTES WITH TENSORING BY A FINITE MODULE OR A PSEUDO-COHERENT COMPLEX

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The naive completion of a Noetherian ring agrees with the derived completion. invoked at proof lines 27371-27373. Identifies termwise completion of the free resolution with the derived completion of K.
- When the derived completion of a module equals its ordinary completion. invoked at proof lines 27372-27373. Justifies replacing the terms (P^n)^ by the derived completions.
- Representation of a derived tensor product by the total complex of a tensor product of representing complexes. used at proof lines 27364-27366. Supplies the complex whose completion is computed.

Source reference. Original lemma 27318: sources/stacks-more-algebra/pinned-original-source.tex, source lines 27318–27318. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original proposition 27218: sources/stacks-more-algebra/pinned-original-source.tex, source lines 27218–27218. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let I be an ideal in a Noetherian ring A. Let $^\wedge$ denote derived completion with respect to I. Let $K \in D^-(A)$.*

(1) *If M is a finite A-module, then $(K \otimes^{\mathbf{L}}_A M)^\wedge = K^\wedge \otimes^{\mathbf{L}}_A M$.*
(2) *If $L \in D(A)$ is pseudo-coherent, then $(K \otimes^{\mathbf{L}}_A L)^\wedge = K^\wedge \otimes^{\mathbf{L}}_A L$.*

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Proof. Let L be as in (2). We may represent K by a bounded above complex $P^\bullet$ of free A-modules. We may represent L by a bounded above complex $F^\bullet$ of finite free A-modules. Since $\mathrm{Tot}(P^\bullet \otimes_A F^\bullet)$ represents $K \otimes^{\mathbf{L}}_A L$ we see that $(K \otimes^{\mathbf{L}}_A L)^\wedge$ is represented by

$$\mathrm{Tot}((P^\bullet)^\wedge \otimes_A F^\bullet)$$

where $(P^\bullet)^\wedge$ is the complex whose terms are the usual = derived completions $(P^n)^\wedge$, see for example Proposition 27218 and Lemma 27318. This proves (2). Part (1) is a special case of (2). $\square$